



CORTEX MED

AI-Powered Smart Patient Triage & Queue Management Platform

Intelligent Triage. Smarter Queues. Better Care.

Empowering hospitals with AI-driven patient prioritization and real-time queue management.

TEAM

LEFTOVERS



Khandaker Ali Ariyan

Full Stack & AI engineer



Fairuz Lubana Binte Newaz

Frontend & System Design

3

THE PROBLEM

Healthcare systems face critical challenges.

Healthcare facilities face increasing pressure due to rising patient numbers and limited medical resources. Most hospitals still rely on first-come-first-served queues, which can delay treatment for patients with critical conditions.

CURRENT PROBLEMS

- Critical patients often wait behind non-urgent cases.
- Manual triage consumes valuable staff time.
- No intelligent queue prioritization.
- Lack of real-time patient tracking.
- Long waiting times reduce patient satisfaction.
- Doctors receive limited decision support.

IMPACT



Increased emergency risk



Longer waiting times



Lower operational efficiency

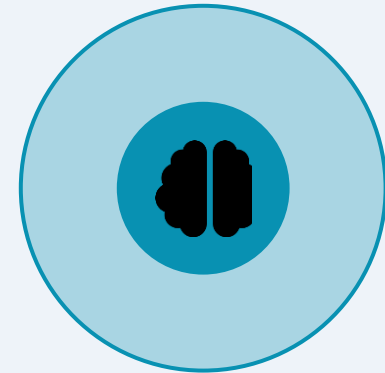


Poor patient experience

An AI-powered smart hospital platform designed to improve patient prioritization, optimize hospital queues, and assist healthcare professionals through intelligent triage.

KEY FEATURES

- AI-powered symptom analysis
- Dynamic patient prioritization
- Intelligent queue management
- Real-time appointment tracking
- Live wait-time estimation
- Analytics dashboard
- Explainable AI reasoning

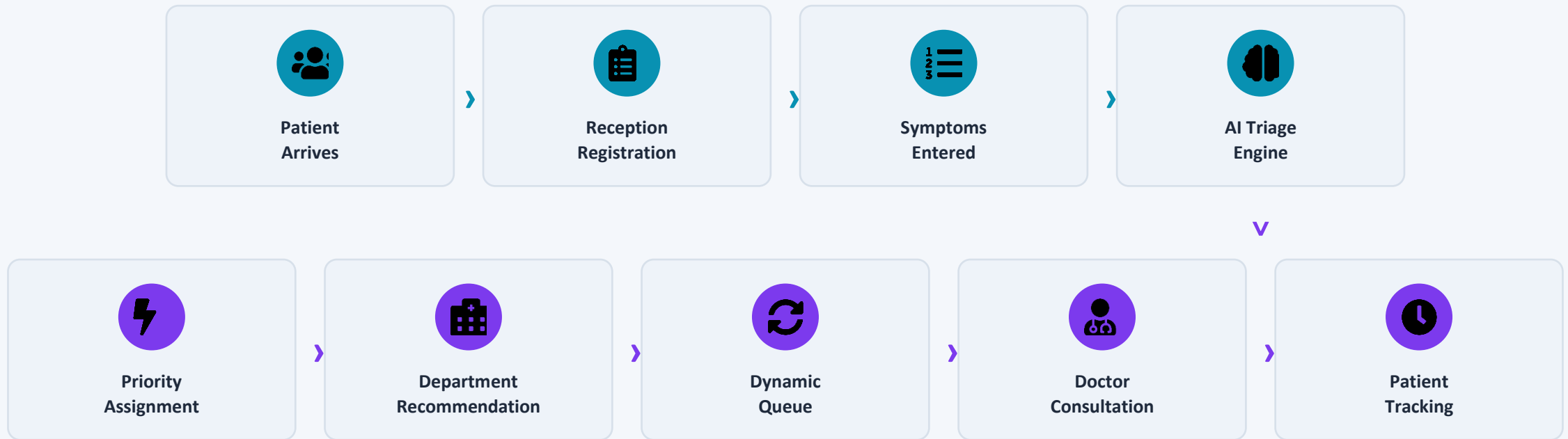


AI ENGINE

*Optimizing patient prioritization
and hospital workflow in real time.*

PROJECT WORKFLOW

End-to-end workflow, from arrival to consultation.



Receptionists register patients and enter symptoms into the system. The AI analyzes the symptoms, estimates urgency, recommends the appropriate department, and assigns a priority level. The queue updates automatically so doctors always see the most urgent patients first. Patients can also monitor their queue position and estimated waiting time in real time.



FRONTEND

Next.js · React · TypeScript



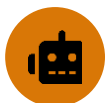
BACKEND

Express.js · Node.js · REST API



DATABASE

MongoDB Atlas



AI SERVICE

FastAPI



LLM

AMD GPU Infrastructure

COMMUNICATION



REST APIs

Standard request/response between client and server.



Socket.IO

Real-time push updates for queue and wait-time changes.



JWT Authentication

Secure, role-based access across every service.

A clean service boundary means the AI backend can be swapped or scaled without touching the frontend.



INPUTS

- Patient symptoms
- Age
- Medical history (optional)



AI PROCESSING

- Symptom understanding
- Medical urgency estimation
- Risk classification
- Department recommendation



OUTPUT

- Priority Level (P1–P4)
- Risk Level
- Confidence Score
- AI Explanation
- Recommended Department

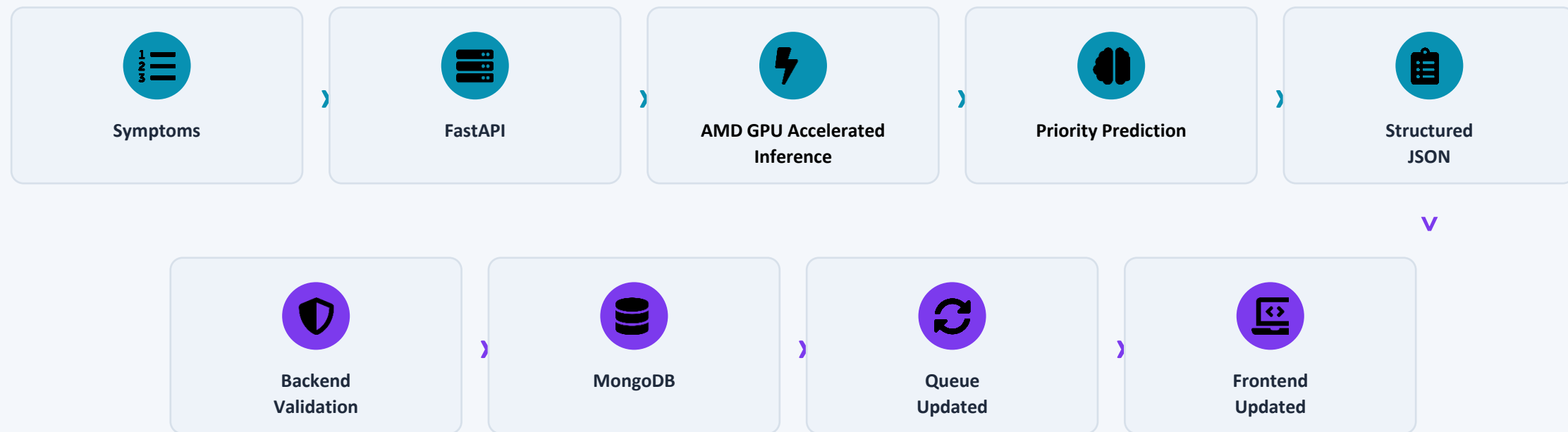
EXAMPLE OUTPUT

Priority: **P1** Risk: **High** Department: **Cardiology** Confidence: **96%**

Reason: Possible cardiac emergency requiring immediate medical attention.

The AI assists healthcare staff by estimating urgency based on symptoms. It does not diagnose diseases and is intended to support — not replace — clinical decision-making.

AI processing pipeline, symptom to synced queue.





RECEPTIONIST

- Register patients
- Book appointments
- AI triage
- Queue monitoring



DOCTOR

- Smart queue
- Call next patient
- Consultation management
- Patient history



ADMIN

- Dashboard
- Analytics
- Staff management
- Hospital settings
- AI monitoring



PATIENT

- Appointment tracking
- Queue status
- Wait-time estimation
- Live updates



FRONTEND

- Next.js 15
- React 19
- TypeScript
- Tailwind CSS
- Zustand
- TanStack Query



BACKEND

- Express.js
- Node.js
- MongoDB Atlas
- JWT
- Socket.IO



AI

- FastAPI
- AI Infrastructure
- AMD GPU
- Pydantic



DEPLOYMENT

- Render
- Vercel
- MongoDB Atlas
- GitHub

TRADITIONAL SYSTEMS

- Static queues
- Manual prioritization
- Limited transparency
- No AI assistance
- Delayed decision making

VS

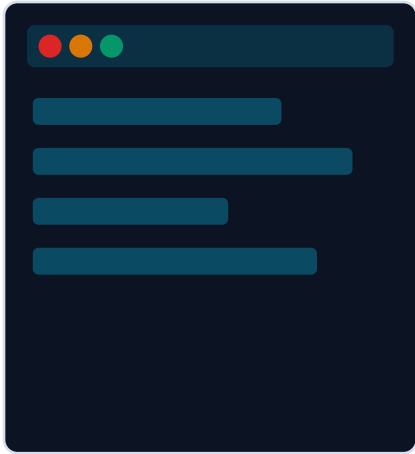
CORTEXMED

- AI-assisted urgency estimation
- Dynamic queue updates
- Live patient tracking
- Explainable AI output
- Real-time synchronization

CortexMed brings intelligence, transparency, and efficiency to hospital operations — instead of replacing doctors, it assists them by intelligently prioritizing patients based on symptom severity.

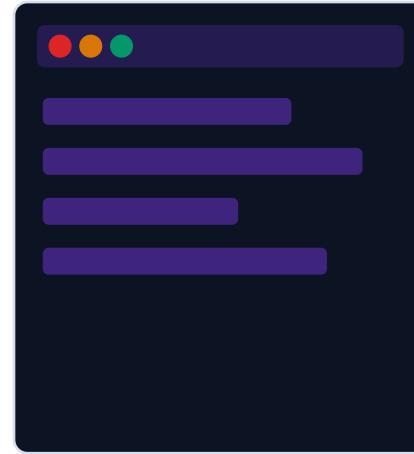
APPLICATION OVERVIEW

A dashboard for every role in the hospital.



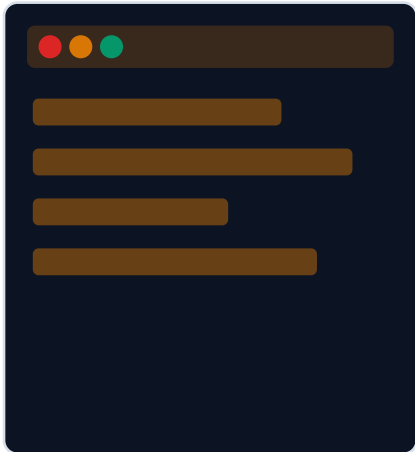
Reception Dashboard

- Register new patients
- Appointment management



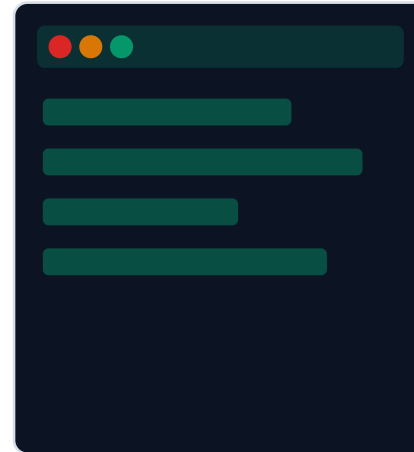
Doctor Dashboard

- Smart queue
- AI reasoning
- Consultation workflow



Patient Tracking

- Queue position
- Estimated waiting time
- Appointment status



Admin Dashboard

- Analytics
- Staff management
- AI engine monitoring
- Hospital performance

Built today for impact tomorrow.

FUTURE FEATURES

- Multi language support
- Voice-based symptom input
- EMR (Electronic Medical Record) integration
- Multi-hospital deployment
- Predictive analytics
- Bed availability optimization
- Ambulance integration
- Wearable health device integration



LONG-TERM VISION

To build a nationwide AI-powered healthcare platform capable of assisting hospitals in delivering faster, smarter, and more efficient patient care.

Built today for impact tomorrow.

A win-win for everyone in the healthcare ecosystem.



FOR PATIENTS

- Reduced waiting time
- Transparent queue tracking
- Better emergency response
- Improved healthcare experience



FOR DOCTORS

- Prioritized patient queues
- Reduced administrative workload
- Faster decision-making
- Better resource utilization



FOR HOSPITALS

- Improved operational efficiency
- Increased patient satisfaction
- Data-driven management
- Smarter healthcare delivery

CortexMed creates a win-win for everyone in the healthcare ecosystem.

1

Receptionist logs in.

2

Registers a new patient.

3

Symptoms are entered.

4

AI analyzes the symptoms.

5

Priority level is assigned.

6

Doctor receives the patient in the smart queue.

7

Consultation is completed.

8

Queue updates automatically.

9

Patient tracking page reflects the latest status.

10

Admin dashboard updates analytics in real time.



CORTEXMED

AI-Powered Smart Patient Triage & Queue Management Platform

WHAT WE BUILT

- AI-assisted symptom analysis
- Smart patient prioritization
- Dynamic queue management
- Real-time hospital coordination
- Explainable AI
- Secure microservice architecture

OUR VISION

To empower hospitals with responsible AI that improves patient care, reduces waiting time, and enhances operational efficiency while keeping healthcare professionals in control of all clinical decisions.

Thank You!

Questions?

TEAM LEFTOVERS

Making hospital triage smarter with AI — for every patient, every department, every day.